

Cosmic Time, Cosmic Entropy and Cosmic Causality

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Abstract:

This article presents a cosmological model that regards the universe as a whole system. From this perspective, each transformation of the cosmic state constitutes a fundamental measure of time, defined as "cosmic time" or "global time". At the same time, the physical process of cosmic evolution is characterized by the tendency of energy distribution to become more uniform in space — a process that corresponds to the increase of entropy. In this way, a mirror-like relationship is established between cosmic time and cosmic entropy: the direction of time's flow is determined by the direction of entropy increase, while entropy changes are driven by the dynamic process of energy conduction and distribution.

The model further suggests that even in the vacuum, mechanisms of energy conduction remain active, allowing entropy to continuously increase. This supports the universality of the Second Law of Thermodynamics at the cosmic scale. The propagation of electromagnetic and gravitational waves through the vacuum is interpreted as energy conduction through the SEQ (Space Elementary Quanta) network, further validating the effectiveness of entropy increase in vacuum conditions.

By integrating thermodynamic time with cosmic time within a causal evolutionary framework, this model provides a new theoretical foundation for understanding the global evolution of the universe, the nature of time, and the universality of entropy.

This article is an introduction of the full cosmological and quantum network model:

"Time-Entropy Mapping; Mass-Gravity Duality; Metric-Frequency Mirroring. Preprints".

<https://doi.org/10.20944/preprints202505.0270.v6>; for the complete theoretical framework, please refer to the updated Preprint Version.

I. The Definition of Cosmic Time: From Global Time to Local Time

In this model, cosmic time (or global time) is defined as the sequence of transformations that the universe as a whole undergoes. Each transformation corresponds to a specific cosmic state. These transformations are not random, but follow the Second Law of Thermodynamics — specifically, the principle of entropy increase. The arrow of time is determined by the direction of increasing entropy, and each cosmic state transition corresponds to a unique entropy value. Since entropy increases in a non-uniform manner, only certain time values actually occur, leading to the discreteness and finiteness of time.

In addition, local time is introduced as an operational concept, used to describe physical processes within specific spatial regions. Local time aligns with the concept of proper time in relativity. However, within the framework of global time, local time measurements always correspond to transformation

processes within specific regions of space.

II. The Principle of Entropy Increase: Uniformization of Energy Distribution and Irreversibility

In this model, entropy is interpreted as a measure of energy distribution. The evolution of the universe is essentially a process in which energy distribution becomes increasingly uniform. In each cosmic transformation, energy flows from high-energy states to low-energy states, eventually tending toward a state of maximum entropy. This energy transfer mechanism ensures the irreversible growth of entropy, even as the universe undergoes expansion or contraction.

Importantly, even in a vacuum, energy transfer and entropy increase still occur. This is because, in this model, the vacuum is not truly empty, but is composed of SEQ in a quantum state. These SEQ possess ground-state energy and are dynamically connected through "spring-like" interactions. When energy is transferred between SEQ, even in the absence of matter, the distribution of energy tends toward uniformity, leading to an increase in entropy.

From this perspective, electromagnetic waves and gravitational waves propagating through a vacuum do not carry energy in isolation; rather, their energy is conducted through the SEQ network. This conduction mechanism implies that even in regions devoid of matter, energy is dynamically exchanged between SEQ, resulting in an increase in entropy. This provides a microscopic explanation for the validity of the Second Law of Thermodynamics in the vacuum — an important extension to classical thermodynamics.

III. The Evolution of Cosmic Entropy: From a Low-Entropy Initial State to Thermal Equilibrium

According to this model, prior to the Big Bang, the universe existed in a highly compressed state. Elastic potential energy was divided into two components: spatial compression potential energy and spatial torsion potential energy. The latter was concentrated in a very small region, forming the initial low-entropy state of the universe. After the Big Bang, this torsion potential energy was released and transformed into the resonance kinetic energy and spin kinetic energy of SEQ, spreading throughout the universe via conduction. This process is a concrete manifestation of entropy increase.

As the universe evolved, energy distribution gradually became more uniform, and entropy continued to rise, eventually approaching a state of maximum entropy. During this process, the universe went through several evolutionary stages, including the homogenization phase and the equilibrium oscillation phase. In the equilibrium oscillation phase, the entropy of the universe approaches its maximum value, and the universe is in a state of thermal equilibrium. However, the trend of entropy increase continues until the maximum entropy value is reached.

Notably, electromagnetic and gravitational waves, as important forms of energy propagation, also contribute to the evolution of cosmic entropy. Their energy is conducted through the SEQ network, even in the vacuum, further driving the increase in entropy. This conduction mechanism not only explains the phenomenon of energy propagation in the vacuum, but also reveals the microscopic basis for the continuous increase of entropy in the vacuum — thereby offering a new theoretical foundation

for the applicability of thermodynamics in vacuum conditions.

IV. The Unification of Cosmic Time and Thermodynamic Time

In this model, the passage of cosmic time is synchronized with the increase of entropy. Each cosmic transformation corresponds to a specific entropy value, and the direction of entropy increase defines the arrow of time. This definition unifies cosmic time and thermodynamic time within a single framework, making the irreversibility of time an intrinsic property of cosmic evolution.

Furthermore, the model acknowledges that although the paths of least action and maximum entropy are not unique, and the rate of entropy increase has some degree of freedom, these do not affect the unidirectionality of time. Therefore, despite the uncertainty of the future, the arrow of time always points in the direction of increasing entropy.

V. A Causal Network from Quantum Microscopic to Cosmic Macroscopic

This model is not only applicable to the evolution of the macroscopic universe but also to the description of the quantum microscopic world. Different energy excitation states of SEQs correspond to different field quanta and fundamental particles, and the mechanisms of energy conduction and distribution remain consistent across both microscopic and macroscopic scales. Therefore, the arrow of thermodynamic time applies not only at the cosmic level but also at the quantum level.

By integrating the principle of entropy increase, the principle of least action, and the principle of maximum entropy, the model constructs a causal network that spans the quantum microscopic and cosmic macroscopic realms. This network not only explains the direction of cosmic evolution but also provides a new perspective on the nature of time.

Conclusion:

In summary, this cosmic model unifies cosmic time and thermodynamic time within a single framework by means of the dynamic structure of SEQ and the mechanism of energy conduction. Within this framework, the principle of entropy increase becomes the fundamental reason for the arrow of time, and the process of energy distribution uniformization ensures the irreversibility of time. Whether in the vacuum or in matter-rich environments, the increase in entropy is always present, making thermodynamic time an intrinsic property of cosmic evolution.

In particular, when electromagnetic or gravitational waves propagate through the vacuum, their energy is conducted through the SEQ network, further driving the evolution of cosmic entropy. This mechanism not only explains the phenomenon of energy propagation in the vacuum, but also reveals the microscopic basis for the continuous increase of entropy in the vacuum — thereby offering a new theoretical foundation for the applicability of thermodynamics in vacuum conditions.

This perspective not only deepens our understanding of cosmic time and entropy, but also lays a theoretical foundation for future exploration of the ultimate fate of the universe.